

Sainik Khaddar

Software Engineer

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SUMMARY

Software Engineer with 2.5+ years of building scalable backend systems and cloud-native applications in C# / .NET, ASP.NET Core, TypeScript, and React. Shipped microservices, RESTful APIs, and event-driven serverless platforms (Azure Functions) powering e-commerce systems **serving 3,500+ retailers at ₹8–10 Cr monthly GMV** — driving a **22% lift in net-new signups, ₹1.4 Cr+ in incremental GMV** via a referral & loyalty subsystem, and an **18% reduction in customer-acquisition cost**.

Applies clean layered design, the repository pattern, dependency injection, and idempotent processing to build maintainable, high-throughput systems. Skilled in CI/CD and containerized deployment (Azure DevOps, Docker, Kubernetes, Prometheus/Grafana) and integration across the Azure stack and third-party systems (SAP, Salesforce, payment gateways) for secure, reliable delivery.

TECHNICAL SKILLS

Languages: C# / .NET, TypeScript, JavaScript (ES6+), SQL

Frameworks & Libraries: ASP.NET Core, .NET 6/8, Entity Framework Core, Dapper, Azure Functions, ReactJS (18), React Native, Redux Toolkit, TanStack React Query, MUI / MUI X DataGrid

Architecture: Microservices, RESTful APIs, Event-Driven Architecture, API Gateway (Ocelot), Repository / Clean Layered Architecture, Dependency Injection, Middleware, Background Services, Serverless, OOP

Cloud & DevOps: Microsoft Azure (SQL, Service Bus, Blob, Functions, Cognitive Search, App Insights), Azure DevOps, Docker, Kubernetes (Helm / k3d), Prometheus, Grafana, AWS

CI/CD: Azure Pipelines (YAML), ARM Templates (IaC), Helm, Git / GitHub

Databases & Caching: SQL Server / Azure SQL, PostgreSQL, Redis (IDistributedCache), Query Optimization & Indexing

Integrations & Security: OAuth 2.0, JWT, SHA512, PayU, Twilio OTP, Firebase Cloud Messaging, WATI WhatsApp API, SAP Ariba / ECC, Icertis CLM, Salesforce REST API, MS Dynamics 365

Methodologies: Agile / Scrum, CI/CD, TDD

PROFESSIONAL EXPERIENCE

Lead Software Engineer (Associate) — PwC India, Kolkata May 2024 – Present

Satya-Retail B2B — FMCG E-Commerce Platform

- **Microservices backend** — Engineered features across a 13-service ASP.NET Core backend (Users, Search, Cart-and-Order, Payment, Subscriptions, Picker, Driver, etc.) sharing 110+ EF Core code-first entities over a common B2BApp.Persistence layer; standardized the Business→Data repository pattern, AutoMapper DTO mapping and middleware-based version gating + central exception logging — cutting feature-delivery cycle time ~30% and production MTTR ~40% across services serving 3,500+ retailers and ₹8–10 Cr monthly GMV.
- **Distributor Ops Console** — Architected a React 18 / TypeScript console (MUI X DataGrid, TanStack Query, Redux Toolkit, react-hook-form) for 300+ distributors and ~50 supervisors running ~8,000 orders/day across 15 warehouses; built a server-paginated Driver-Planning grid with DriverToBeat/DriverToFOS toggle, multi-driver selection, dnd-kit drag-and-drop beat reassignment, and a “Freeze Plan” action batch-triggering picker tasks across 2,500+ orders in one click — collapsing a 3–4 hour Excel dispatch into under 5 minutes and cutting picker idle time ~40%. Added cascading filters (warehouse→beat→FOS→delivery-type), split FMCG/Non-FMCG assignment, and placedQty-vs-packedQty reconciliation — reducing pick-pack errors ~65% and lifting throughput ~35%.
- **Referral & Loyalty subsystem** — Designed an end-to-end referral & loyalty system spanning the Users API and a .NET 8 Azure Function — added 2 EF Core entities with migrations, generated cryptographically-strong 8-char codes via RandomNumberGenerator over a 32-char ambiguity-free alphabet with self-referral guards, and wired into the ASP.NET Identity + JWT/Twilio OTP flow; a TimerTrigger function batches transactions and syncs loyalty points to Dynamics 365 via an Azure AD OAuth 2.0 client-credentials ErpAuthService with in-memory token caching. Drove a 22% lift in net-new signups, ₹1.4 Cr+ incremental GMV, ~18% lower paid CAC, a 99.6% ERP sync rate, and ~15% higher repeat-order frequency.
- **SO→TO & GST E-Way Bill platform** — Consolidated four fragmented dispatch operations (Check TO, Retry TO, E-Way Bill, Reco-Mismatch) into a single trip-scoped tabbed modal with idempotent localStorage-backed generation state and one-click E-Way Bill + Delivery Challan PDF export — making the platform GST-compliant for ₹4 Cr+/month of inter-warehouse movement, eliminating ~20 clicks/dispatch (~40 hrs/week saved), and recovering ~₹12 lakh/month in previously-leaked inventory via a debounced reco-mismatch sub-system.
- **Search optimization** — Re-architected the Search microservice (Azure Cognitive Search + Redis IDistributedCache) from naive string-concat keys to versioned, multi-dimensional keys (search:v2:{term}:{pincode}:{userId}:{languageId}:{isFos}) eliminating cross-tenant/locale collisions, added barcode/item-number lookup, prevented empty-result cache poisoning, and merged real-time inventory-on-hand + warehouse-multiplier data — cutting p95 latency 63% (850ms → 310ms), raising cache hit rate 58% → 84%, and lifting search-to-cart conversion ~9% (≈₹60–75 L/month incremental order value).
- **Mobile delivery & CI/CD** — Built the React Native 0.80 / TypeScript consumer app’s data layer — Redux Toolkit (9 feature slices), TanStack Query (50+ keys, 112+ hooks) over an Axios client with JWT interceptors and a queued 401 refresh-token retry pipeline, plus Branch.io deep linking for referral redemption. Owned release delivery to the Play Store and App Store, including a full 4KB→16KB memory-page-alignment migration for Google Play compliance and BrowserStack device validation; built CI/CD pipelines publishing frontend/backend as Docker images to Azure Container Registry and deploying to Azure.

Le-Marche D2C — FMCG E-Commerce Platform

- **Microservices platform** — Architected a distributed .NET 6/8 platform (12+ services — Cart, Payment, Catalog, Subscriptions, Users, Search behind an Ocelot API Gateway with JWT auth and role-based authorization) supporting ~8,000 daily orders and ₹6 Cr+ monthly GMV at 99.9% uptime, absorbing 3× festive-season traffic with zero downtime; bulk CSV pipelines (CsvHelper, EFCore.BulkExtensions, Azure Blob persistence, row-level error tracking) processed 50,000+ rows in under 90s — cutting distributor onboarding from 6 hours to 15 minutes and scaling the network from ~120 to 300+ distributors in under a year.
- **Payments, subscriptions & serviceability** — Led the team that eliminated payment failures end-to-end — integrated PayU (SHA512 hashing, SURL/FURL callbacks) and an idempotent order-to-payment state machine resolving duplicate-cart race conditions, lifting success rate from 91% to 97.5% (≈₹10 L/month recovered); built the recurring-subscription experience (create/edit/pause/slot-change) growing subscribers to ~12,000 and ₹1.8 Cr+ monthly recurring GMV, and a geolocation + pin-code serviceability layer (Google Maps SDK, AsyncStorage-cached addresses) that cut undeliverable orders ~75% and refunds ~₹8 L/month.
- **Performance & notifications** — Tuned SQL Server + EF Core data access across 97 repositories and 20+ entities (projection queries, async streaming, indexed reads, tuned command timeouts) — cutting API response time 63% (850ms → 310ms) and DB CPU 35% at peak, deferring a database-tier upgrade (≈₹1.5 L/month saved); built a notification hub (Firebase Cloud Messaging + WATI WhatsApp) orchestrating ~400K messages/month and lifting cart-abandonment recovery 18% (≈₹25 L/month recovered).

Syngene Internationals — Pharmaceutical Client

- **Contract-lifecycle middleware** — Designed an event-driven middleware on Azure Functions (.NET 8 isolated worker, 9 Timer/HTTP/Service Bus triggers, 3 decoupled Service Bus queues with at-least-once delivery and 5× exponential retry) automating create/amend/terminate flows across SAP Ariba, Icertis CLM, and SAP ECC for a CRO managing ~₹4,500 Cr annual procurement — cutting contract turnaround from ~7 days to under 2 hours and eliminating ~40 hrs/week of manual entry.
- **Integration layer & data sync** — Built HttpClientFactory + Polly REST clients across 3 auth schemes (OAuth2 client-credentials, Azure AD bearer, Basic for ECC OData) with App Insights distributed tracing and strict parent-agreement validation, driving duplicate-posting errors down ~92% (protecting ~1,200 monthly events worth ~₹375 Cr/month); engineered a Dapper-based quote-to-order sync across SAP, Azure SQL, and Salesforce REST with idempotent upserts and SQL row/update-lock concurrency for exactly-once processing — handling \$120M+ annual revenue and 500K+ records/day at 99.9% accuracy.
- **Cloud delivery & observability** — Owned CI/CD on Azure with Azure DevOps YAML pipelines, zip-deploy, and ARM-templated infrastructure (Functions, Azure SQL, Storage, App Insights) for repeatable provisioning, with Azurite-emulated local Service Bus/Blob/Queue testing; instrumented a status-driven processing state machine and centralized audit logging — cutting mean-time-to-detect on failed syncs ~60% and supporting 30% YoY transaction-volume growth with zero added operational overhead.

DevOps Engineer Intern — Bespoke Labs-AI, California (Remote) *Apr 2023 – May 2023*

- Deployed a FastAPI + PostgreSQL microservice stack with Prometheus/Grafana observability via Helm on Kubernetes, and a full Docker-in-Docker (k3d) setup where a single image spins up a cluster, installs Helm charts, and exposes all endpoints — a reproducible, production-grade local DevOps environment.

KEY ACHIEVEMENTS

- **Client recognition** — Received written recognition from the DGM Product at Satya Retail for shipped features (referral program, carousel “view all,” floating cart), cited as having proven valuable for the business — a direct product-owner endorsement of revenue-impacting work.
- **Enterprise-scale automation** — Automated a contract-lifecycle platform managing ₹4,500 Cr in annual procurement spend, collapsing turnaround from ~7 days to under 2 hours and reducing posting errors ~92%.
- **Team Lead for urgent Production incident fix** — Led the team that eliminated payment failures end-to-end, lifting success rate from 91% to 97.5% and recovering ~₹10 L/month in previously failed transactions — drove the PayU gateway integration (SHA512 hashing, SURL/FURL callbacks) and architected an idempotent order-to-payment state machine that fully resolved the duplicate-cart race conditions plaguing checkout.

EDUCATION & CERTIFICATIONS

Jadavpur University — B.E. in Electrical Engineering (CGPA 8.33/10) *Jan 2020 – May 2024*

Certifications: Microsoft Azure AZ-400 (DevOps Engineer Expert), AZ-204 (Developer Associate)